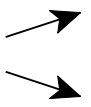


Section 5.2/5.3

- I. Summary: Algebra, Limits, Calculus  Differential
Integral
- II. Integral Calculus

- A. Particular Example (Uniform Partition): Computation of Area as a Sum
B. General Setting: Construction of the Riemann Integral

III. General Scheme for $y = f(x)$ on the Interval $[a, b]$

- A. Partition the Interval

$$\text{Uniform Partition } (\|P\| = \frac{b-a}{n})$$

$$P = \{a = x_0 < a + \Delta x = x_1 < a + 2\Delta x = x_2 < a + 3\Delta x = x_3 < \cdots a + n\Delta x = x_n = b\}$$

$$\Delta x = \frac{b-a}{n}$$

$$\text{General Setting } (\|P\| = \max_{1 \leq k \leq n} \Delta x_k)$$

$$P = \{a = x_0 < x_1 < x_2 < \cdots < x_n = b\}$$

$$\Delta x_k = x_k - x_{k-1}$$

- B. Choose a Distinguished Point x_k^* in each Subinterval of the Partition

$$\text{Uniform Partition } x_k^* = a + k\Delta x$$

$$\text{General Setting } x_k^* \in [x_{k-1}, x_k]$$

- C. Compute Function Values $f(x_k^*)$ over the Partition

- D. Form the Riemann Sum of the Products $f(x_k^*)\Delta x_k$ over the Partition

Uniform Partition

$$f(a + \Delta x)\Delta x + f(a + 2\Delta x)\Delta x + f(a + 3\Delta x)\Delta x + \cdots + f(a + n\Delta x)\Delta x$$

General Setting

$$f(x_1^*)\Delta x_1 + f(x_2^*)\Delta x_2 + f(x_3^*)\Delta x_3 + \cdots + f(x_n^*)\Delta x_n$$

Examples (Graphical)

IV. Summation Notation

A. Basic Properties

- | | |
|-------------------------|-------------------|
| 1. Constant Term | 2. Sum Rule |
| 3. Scalar Multiple Rule | 4. Linearity Rule |
| 5. Subtotal Rule | 6. Dominance Rule |

1. Formulas

2. Examples

V. Riemann Integral for $y = f(x)$ on the Interval $[a, b]$

A. Form the Riemann Sum as Described in Step III Above

B. Compute the Limit of the Riemann Sum as $\|P\| \rightarrow 0$

1. ***If*** the limit exists, we say the function f is integrable on the interval $[a, b]$

2. We denote the limit by $\int_a^b f(x)dx$ and call it the definite integral of f from a to b

3. The function f is called the integrand; the interval $[a, b]$ is called the interval of integration; the endpoints a and b are called the lower and upper limits of integration, resp.

4. Theorem (Advanced Calculus) *If f is continuous on an interval $[a, b]$, then f is integrable on $[a, b]$.*

5. If f is continuous on $[a, b]$ and if $f(x) \geq 0$ on $[a, b]$, then $\int_a^b f(x)dx = \text{area under the curve } y = f(x) \text{ over the interval } [a, b]$.

Examples

C. Properties of the Definite Integral

- | | |
|------------------------|--|
| 1. Integral at a Point | 2. Interchanging the Limits of Integration |
| 3. Linearity | 4. Subdivision Rule |
| 5. Dominance | |

Examples